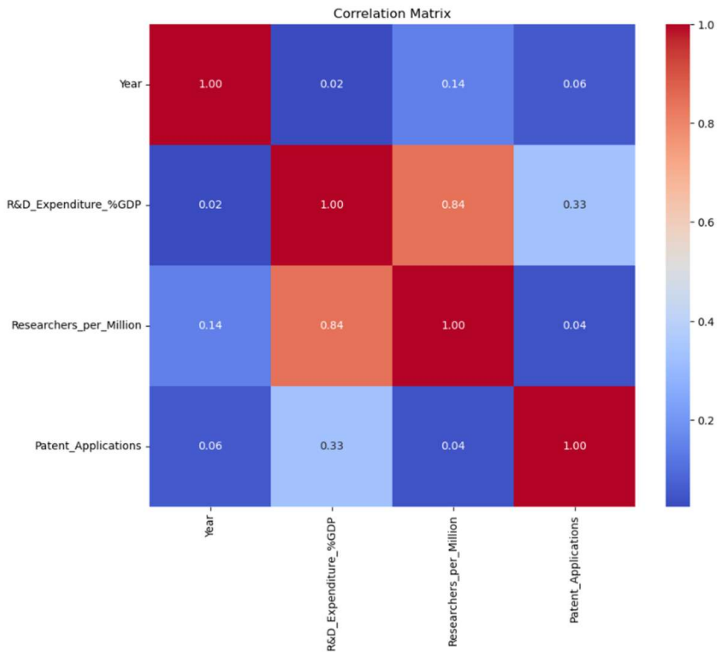
Plotting and Results

1.Data Visualization and Plotting



A heatmap was generated using Seaborn to understand the correlation between numerical variables.

Observation:

- **R&D Expenditure (%GDP)** shows positive correlation with **Patent Applications**.
- **Researchers per Million** also shows strong positive relationship.
- Both independent variables contribute significantly toward innovation output.

This confirms that higher research investment and human capital increase patent production.

2. Model Results and Performance Analysis

Two regression models were implemented:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into training (80%) and testing (20%) sets.

2.1 Feature Importance (Random Forest)

From SDG-9 Impact Analysis:

- **R&D Expenditure Importance: 0.323**
- **Researchers per Million Importance: 0.677**

Interpretation:

- Researchers per Million contributes **67.7% influence**
- R&D Expenditure contributes **32.3% influence**

This indicates that skilled human capital plays a dominant role in innovation output.

2.2 Model Performance Metrics

Linear Regression

- **R^2 Score: 0.345**
- **RMSE: 165,027.809**

Interpretation:

- Model explains only **34.5% variation**
- Higher RMSE indicates large prediction errors
- Linear model fails to capture complex relationships

Random Forest Regression

- **R^2 Score: 0.775**
- **RMSE: 96,740.615**

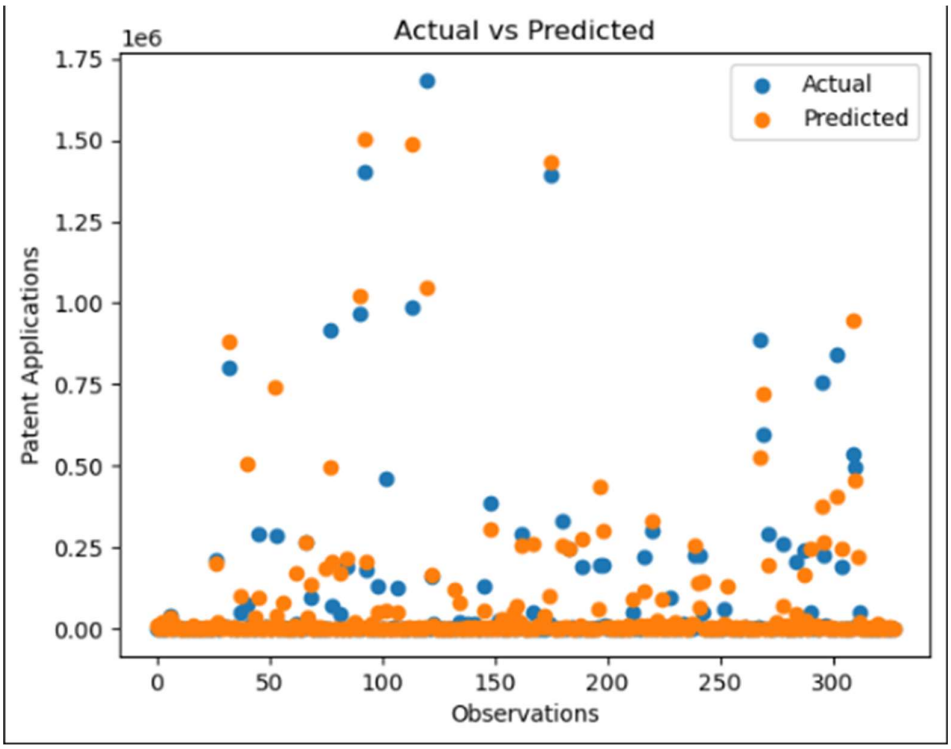
Interpretation:

- Model explains **77.5% variation in patent output**
- Much lower RMSE compared to Linear Regression
- Captures nonlinear patterns effectively

2.3 Model Comparison

Model	R^2 Score	RMSE	Performance
Linear Regression	0.345	165,027	Moderate
Random Forest	0.775	96,740	Strong

2.4 Actual vs Predicted Visualization



The scatter plot compares:

- Blue points → Actual Patent Applications
- Orange points → Predicted Patent Applications

Observations:

- Predictions are closer to actual values in Random Forest.
- Some variation exists at higher patent values.
- Model handles medium and low patent ranges accurately.
- Slight deviation for extreme values.

This confirms strong predictive capability.